

Activity_Explore hypothesis testing

May 26, 2026

1 Activity: Explore hypothesis testing

1.1 Introduction

You work for an environmental think tank called Repair Our Air (ROA). ROA is formulating policy recommendations to improve the air quality in America, using the Environmental Protection Agency's Air Quality Index (AQI) to guide their decision making. An AQI value close to 0 signals “little to no” public health concern, while higher values are associated with increased risk to public health.

They've tasked you with leveraging AQI data to help them prioritize their strategy for improving air quality in America.

ROA is considering the following decisions. For each, construct a hypothesis test and an accompanying visualization, using your results of that test to make a recommendation:

1. ROA is considering a metropolitan-focused approach. Within California, they want to know if the mean AQI in Los Angeles County is statistically different from the rest of California.
2. With limited resources, ROA has to choose between New York and Ohio for their next regional office. Does New York have a lower AQI than Ohio?
3. A new policy will affect those states with a mean AQI of 10 or greater. Will Michigan be affected by this new policy?

Notes: 1. For your analysis, you'll default to a 5% level of significance. 2. Throughout the lab, for two-sample t-tests, use Welch's t-test (i.e., setting the `equal_var` parameter to `False` in `scipy.stats.ttest_ind()`). This will account for the possibly unequal variances between the two groups in the comparison.

1.2 Step 1: Imports

To proceed with your analysis, import `pandas` and `numpy`. To conduct your hypothesis testing, import `stats` from `scipy`.

Import Packages

```
[1]: # Import relevant packages

    ### YOUR CODE HERE ###
```

```
import pandas as pd
import numpy as np
from scipy import stats
```

You are also provided with a dataset with national Air Quality Index (AQI) measurements by state over time for this analysis. `Pandas` was used to import the file `c4_epa_air_quality.csv` as a dataframe named `aqi`. As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the `.csv` file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

Note: For purposes of your analysis, you can assume this data is randomly sampled from a larger population.

Load Dataset

```
[2]: # RUN THIS CELL TO IMPORT YOUR DATA.

### YOUR CODE HERE ###

aqi = pd.read_csv('c4_epa_air_quality.csv')
```

1.3 Step 2: Data Exploration

1.3.1 Before proceeding to your deliverables, explore your datasets.

Use the following space to surface descriptive statistics about your data. In particular, explore whether you believe the research questions you were given are readily answerable with this data.

```
[3]: # Explore your dataframe `aqi` here:

### YOUR CODE HERE ###

print("Using head() to show a sample of data")
print(aqi.head())
print("Using describe() to summarize AQI")
print(aqi.describe(include='all'))
```

Using head() to show a sample of data

	Unnamed: 0	date_local	state_name	county_name	city_name \
0	0	2018-01-01	Arizona	Maricopa	Buckeye
1	1	2018-01-01	Ohio	Belmont	Shadyside
2	2	2018-01-01	Wyoming	Teton	Not in a city
3	3	2018-01-01	Pennsylvania	Philadelphia	Philadelphia
4	4	2018-01-01	Iowa	Polk	Des Moines

		local_site_name	parameter_name \
0		BUCKEYE	Carbon monoxide

```

1                               Shadyside Carbon monoxide
2 Yellowstone National Park - Old Faithful Snow ... Carbon monoxide
3                               North East Waste (NEW) Carbon monoxide
4                               CARPENTER Carbon monoxide

```

```

      units_of_measure  arithmetic_mean  aqi
0 Parts per million      0.473684      7
1 Parts per million      0.263158      5
2 Parts per million      0.111111      2
3 Parts per million      0.300000      3
4 Parts per million      0.215789      3

```

Using describe() to summarize AQI

```

      Unnamed: 0  date_local  state_name  county_name  city_name \
count  260.000000      260      260      260      260
unique      NaN          1          52      149      190
top      NaN  2018-01-01  California  Los Angeles  Not in a city
freq      NaN          260          66          14          21
mean  129.500000      NaN      NaN      NaN      NaN
std    75.199734      NaN      NaN      NaN      NaN
min     0.000000      NaN      NaN      NaN      NaN
25%    64.750000      NaN      NaN      NaN      NaN
50%   129.500000      NaN      NaN      NaN      NaN
75%   194.250000      NaN      NaN      NaN      NaN
max   259.000000      NaN      NaN      NaN      NaN

```

```

      local_site_name  parameter_name  units_of_measure  arithmetic_mean \
count              257              260              260      260.000000
unique             253              1              1              NaN
top      Kapolei Carbon monoxide  Parts per million              NaN
freq              2              260              260              NaN
mean      NaN      NaN      NaN      NaN      0.403169
std      NaN      NaN      NaN      NaN      0.317902
min      NaN      NaN      NaN      NaN      0.000000
25%      NaN      NaN      NaN      NaN      0.200000
50%      NaN      NaN      NaN      NaN      0.276315
75%      NaN      NaN      NaN      NaN      0.516009
max      NaN      NaN      NaN      NaN      1.921053

```

```

      aqi
count  260.000000
unique      NaN
top      NaN
freq      NaN
mean    6.757692
std     7.061707
min     0.000000
25%     2.000000
50%     5.000000

```

```
75%      9.000000
max      50.000000
```

HINT 1

Consider referring to the material on descriptive statistics.

HINT 2

Consider using `pandas` or `numpy` to explore the `aqi` dataframe.

HINT 3

Any of the following functions may be useful: - `pandas`: `describe()`, `value_counts()`, `shape()`, `head()` - `numpy`: `unique()`, `mean()`

Question 1: From the preceding data exploration, what do you recognize? From the preceding data exploration, we recognize that we have county-level data for the first hypothesis.

1.4 Step 3. Statistical Tests

Before you proceed, recall the following steps for conducting hypothesis testing:

1. Formulate the null hypothesis and the alternative hypothesis.
2. Set the significance level.
3. Determine the appropriate test procedure.
4. Compute the p-value.
5. Draw your conclusion.

1.4.1 Hypothesis 1: ROA is considering a metropolitan-focused approach. Within California, they want to know if the mean AQI in Los Angeles County is statistically different from the rest of California.

Before proceeding with your analysis, it will be helpful to subset the data for your comparison.

```
[6]: # Create dataframes for each sample being compared in your test

### YOUR CODE HERE ###

ca_la = aqi[aqi['county_name']=='Los Angeles']
ca_other = aqi[(aqi['state_name']=='California') & (aqi['county_name']!='Los_
→Angeles')]
```

HINT 1

Consider referencing the material on subsetting dataframes.

HINT 2

Consider creating two dataframes, one for Los Angeles, and one for all other California observations.

HINT 3

For your first dataframe, filter to `county_name` of Los Angeles. For your second dataframe, filter to `state_name` of California and `county_name` not equal to Los Angeles.

Formulate your hypothesis: Formulate your null and alternative hypotheses:

- H_0 : There is no difference in the mean AQI between Los Angeles County and the rest of California.
- H_A : There is a difference in the mean AQI between Los Angeles County and the rest of California.

Set the significance level:

```
[7]: # For this analysis, the significance level is 5%

    ### YOUR CODE HERE

significance_level = 0.05
significance_level
```

[7]: 0.05

Determine the appropriate test procedure: Here, you are comparing the sample means between two independent samples. Therefore, you will utilize a **two-sample -test**.

Compute the P-value

```
[8]: # Compute your p-value here

    ### YOUR CODE HERE ###

stats.ttest_ind(a=ca_la['aqi'], b=ca_other['aqi'], equal_var=False)
```

[8]: Ttest_indResult(statistic=2.1107010796372014, pvalue=0.049839056842410995)

HINT 1

Consider referencing the material on how to perform a two-sample t-test.

HINT 2

In `ttest_ind()`, `a` is the `aqi` column from our “Los Angeles” dataframe, and `b` is the `aqi` column from the “Other California” dataframe.

HINT 3

Be sure to set `equal_var = False`.

Question 2. What is your P-value for hypothesis 1, and what does this indicate for your null hypothesis? The p-value for hypothesis 1 is 0.0498, which is less than the significance level of 0.05. This indicates that we reject the null hypothesis. Meaning, in this case, a metropolitan strategy may make more sense.

1.4.2 Hypothesis 2: With limited resources, ROA has to choose between New York and Ohio for their next regional office. Does New York have a lower AQI than Ohio?

Before proceeding with your analysis, it will be helpful to subset the data for your comparison.

```
[9]: # Create dataframes for each sample being compared in your test

### YOUR CODE HERE ###

ny = aqi[aqi['state_name']=='New York']
ohio = aqi[aqi['state_name']=='Ohio']
```

HINT 1

Consider referencing the materials on subsetting dataframes.

HINT 2

Consider creating two dataframes, one for New York, and one for Ohio observations.

HINT 3

For your first dataframe, filter to `state_name` of New York. For your second dataframe, filter to `state_name` of Ohio.

Formulate your hypothesis: Formulate your null and alternative hypotheses:

- H_0 : The mean AQI of New York is greater than or equal to that of Ohio.
- H_A : The mean AQI of New York is **below** that of Ohio.

Significance Level (remains at 5%)

Determine the appropriate test procedure: Here, you are comparing the sample means between two independent samples in one direction. Therefore, you will utilize a **two-sample -test**.

Compute the P-value

```
[10]: # Compute your p-value here

### YOUR CODE HERE ###
```

```
tstat, pvalue = stats.ttest_ind(a=ny['aqi'], b=ohio['aqi'], alternative='less',
    ↪equal_var=False)
print(tstat)
print(pvalue)
```

```
-2.025951038880333
0.030446502691934697
```

HINT 1

Consider referencing the material on how to perform a two-sample t-test.

HINT 2

In `ttest_ind()`, `a` is the `aqi` column from the “New York” dataframe, and `b` is the `aqi` column from the “Ohio” dataframe.

HINT 3

You can assign `tstat`, `pvalue` to the output of `ttest_ind`. Be sure to include `alternative = less` as part of your code.

Question 3. What is your P-value for hypothesis 2, and what does this indicate for your null hypothesis? The p-value for hypothesis 2 is 0.03, which is less than the significance level of 0.05. This means we reject the null hypothesis. Meaning, at the 5% significance level, New York has a lower mean AQI than Ohio.

1.4.3 Hypothesis 3: A new policy will affect those states with a mean AQI of 10 or greater. Will Michigan be affected by this new policy?

Before proceeding with your analysis, it will be helpful to subset the data for your comparison.

```
[11]: # Create dataframes for each sample being compared in your test

### YOUR CODE HERE ###

michigan = aqi[aqi['state_name']=='Michigan']
```

HINT 1

Consider referencing the material on subsetting dataframes.

HINT 2

Consider creating one dataframe which only includes Michigan.

Formulate your hypothesis: Formulate your null and alternative hypotheses here:

- H_0 : The mean AQI of Michigan is less than or equal to 10.
- H_A : The mean AQI of Michigan is greater than 10.

Significance Level (remains at 5%)

Determine the appropriate test procedure: Here, you are comparing one sample mean relative to a particular value in one direction. Therefore, you will utilize a **one-sample -test**.

Compute the P-value

```
[12]: # Compute your p-value here

### YOUR CODE HERE ###

tstat, pvalue = stats.ttest_1samp(michigan['aqi'], 10, alternative='greater')
print(tstat)
print(pvalue)
```

```
-1.7395913343286131
```

```
0.9399405193140109
```

HINT 1

Consider referencing the material on how to perform a one-sample t-test.

HINT 2

In `ttest_1samp`, you are comparing the `aqi` column from your Michigan data relative to 10, the new policy threshold.

HINT 3

You can assign `tstat`, `pvalue` to the output of `ttest_1samp`. Be sure to include `alternative = greater` as part of your code.

Question 4. What is your P-value for hypothesis 3, and what does this indicate for your null hypothesis? The p-value for hypothesis 3 is 0.940, which is greater than the significance level of 0.05. This indicates that we fail to reject the null hypothesis. Meaning, at the 5% significance level, we cannot conclude that the mean AQI of Michigan is greater than 10. This implies that Michigan would most likely not be affected by the new policy.

1.5 Step 4. Results and Evaluation

Now that you've completed your statistical tests, you can consider your hypotheses and the results you gathered.

Question 5. Did your results show that the AQI in Los Angeles County was statistically different from the rest of California? Yes, the results show that the AQI in Los Angeles County was statistically different from the rest of California.

Question 6. Did New York or Ohio have a lower AQI? At 5% significance level, New York had a lower AQI than Ohio.

Question 7: Will Michigan be affected by the new policy impacting states with a mean AQI of 10 or greater? According to the tests, it is unlikely that Michigan would be affected by the new policy.

2 Conclusion

What are key takeaways from this lab? The following are the key takeaways from this lab: Even with small sample sizes, the variation within the data is enough to allow us to make statistically significant conclusions. We identified at the 5% significance level that the Los Angeles mean AQI was statistically different from the rest of California, and that New York does have a lower mean AQI than Ohio. However, we were unable to conclude at the 5% significance level that Michigan's mean AQI was greater than 10.

What would you consider presenting to your manager as part of your findings? For each test, we would present the null and alternative hypothesis, then describe our conclusion and the resulting p-value that drove that conclusion. We would specify the type of test we chose, whether that tail was one-tail or two-tailed, and how we performed the t-test from stats.

What would you convey to external stakeholders? We would convey the level of significance (5%) and our conclusion, along with providing the sample statistics being compared in each case for more context.

Congratulations! You've completed this lab. However, you may not notice a green check mark next to this item on Coursera's platform. Please continue your progress regardless of the check mark. Just click on the "save" icon at the top of this notebook to ensure your work has been logged.